



Post-Quantum Secure Firmware Distribution and Vehicle Authentication using CRYSTALS-Dilithium

Batch/Group/Team No.: 11

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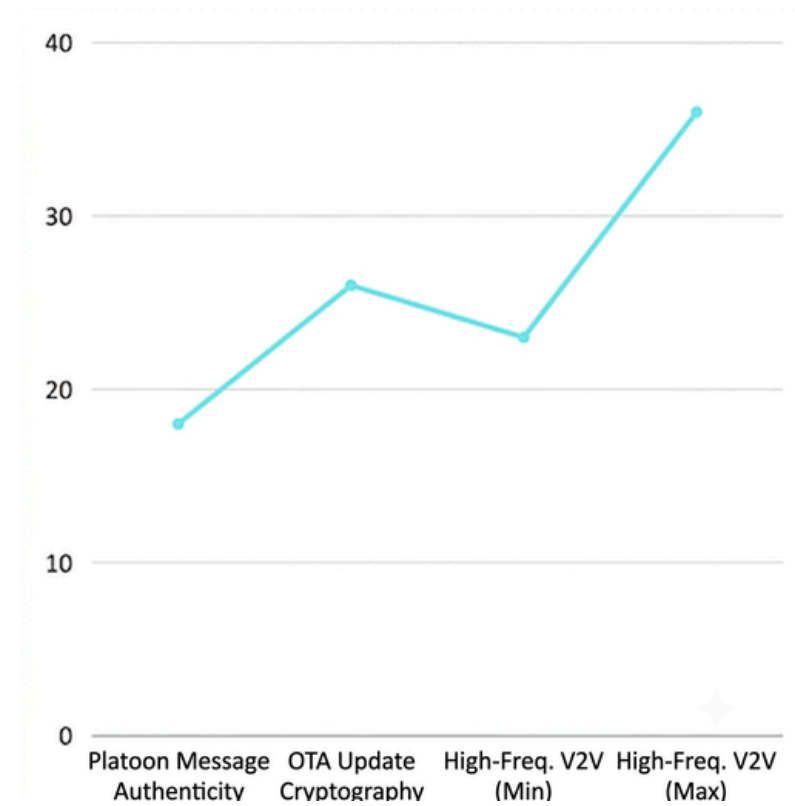
Overview

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Motivation

- A single forged safety message can affect every vehicle within a platoon, message authenticity a critical requirement for safe operation.
- Modern connected vehicles support frequent Over-the-Air (OTA) firmware updates, requiring every update to be cryptographically verified before installation to prevent unauthorized code execution.
- Autonomous vehicle platoons exchange 10–50 authenticated safety messages per second, requiring fast and reliable signature verification.





SDG Alignment & IKS Integration

- **SDG Alignment:**
 - **SDG 9 – Industry, Innovation, and Infrastructure:** Secure and resilient intelligent transportation systems.
 - **SDG 11 – Sustainable Cities and Communities:** Improved road safety and reliable autonomous mobility.
- **Indian Knowledge System (IKS):**
 - **Raksha (Protection) :** Proactive safeguarding of safety-critical transportation systems through quantum-resilient security.
 - **Anvikshiki (Scientific Inquiry) :** Application of analytical reasoning and cryptographic innovation to solve emerging cybersecurity challenges.



Literature Survey (Existing Technology)

Ref.No	Title, Author, Year	Methodology Used	Dataset / Evaluation	Advantages	Disadvantages
[1]	Quantum-resistant Transport Layer Security, García et al., 2024	Hybrid TLS integrating Kyber and Dilithium with modified OpenSSL prototype	Measured handshake latency, overhead, and CPU usage in a controlled network setup	Strong quantum resistance; practical protocol implementation	Higher overhead; not designed for vehicular environments
[2]	On Hardware Acceleration of Quantum-Resistant FOTA, Perazzo, 2024	Experimental implementation on automotive hardware	Firmware signing benchmarks	Real-world OTA feasibility; implementation-focused	Emphasizes signatures more than V2V communication
[3]	Efficient Implementation of CRYSTALS-Kyber on ESP32, 2025	Embedded system implementation	Runtime and memory analysis	Shows PQ feasibility on constrained devices	Hardware-specific results
[4]	Open Quantum Safe: Practical Quantum-Safe Cryptography, Stebila et al., 2020	Library-based implementation and TLS integration	Handshake performance tests	Production-ready crypto framework	Requires integration effort



Literature Survey (Existing Technology)

Ref.No	Title, Author, Year	Methodology Used	Dataset / Evaluation	Advantages	Disadvantages
[5]	CRYSTALS-Kyber: A CCA-Secure Module-Lattice-Based KEM, Bos et al., 2018	Algorithm design with reference implementation	Security and performance benchmarks	NIST-selected; strong security	More theoretical than system-level
[6]	CRYSTALS-Dilithium: Digital Signatures from Module Lattices, Ducas et al., 2018	Signature scheme with implementation notes	Computational analysis	Quantum-resistant authentication	Larger signature sizes
[7]	Post-Quantum Security for IoT Devices, Alkim et al., 2016	Implementation study on embedded platforms	Device-level benchmarks	Demonstrates real-world deployment	Older performance data
[8]	Secure Vehicular Communication Using Cryptographic Techniques: A Survey, Lenka & Rath (India), 2019	Threat and protocol analysis	Literature-based	Highlights vehicular security needs	Does not address PQ cryptography
[9]	3GPP LTE vs IEEE 802.11p for Vehicular Safety, Vinel, 2012	Network simulation	Vehicular traffic scenarios	Identifies low-latency requirements	Focus on classical security



PROBLEM STATEMENT

- Ensuring the authenticity of communicating vehicles and the integrity of safety-critical messages remains a major challenge in autonomous vehicle platoon networks. As quantum computing threatens the security of classical digital signature algorithms, a quantum-resistant authentication mechanism is essential to establish trust and prevent unauthorized communication.

Objective Of The Project

- To develop a post-quantum authentication framework using CRYSTALS-Dilithium that ensures secure vehicle authentication, message integrity, and trusted communication in autonomous vehicle platoon networks.





Proposed System / Methodology

1. Vehicle Registration:

- Generates Dilithium public/private key pairs.
- Shares public keys through the trusted server.

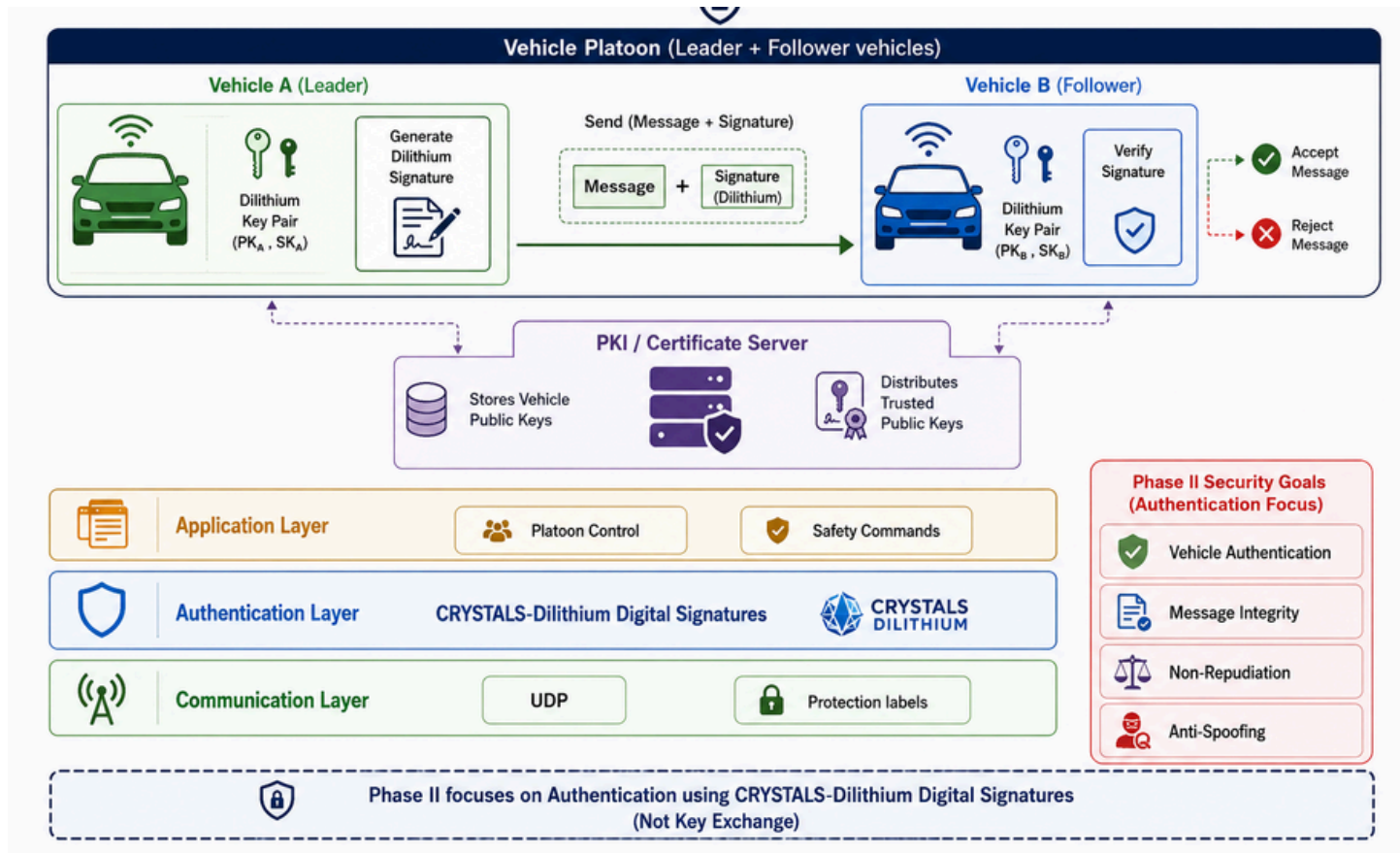
2. Message Authentication:

- Signs outgoing platoon messages using the Dilithium private key.
- Attaches the digital signature to each message.

3. Signature Verification:

- Verifies received signatures using the sender's public key.
- Accepts valid messages and rejects forged or modified messages.

Proposed System / Methodology





Results

```
● PS D:\My Projects\Capstone Project> env\Scripts\Activate
● (env) PS D:\My Projects\Capstone Project> python demo.py mitm

=====
SCENARIO 2: Man-in-the-middle attack (identity checks ON)
=====
Leader tries to reach follower-1 -- but the address is the attacker's:
- [leader->follower-1] sent HELLO + nonce (8a12083c...)
- [leader->follower-1] got follower cert 'mitm' + Kyber public key (1184 B)
[BLOCKED] MITM stopped: attacker's certificate is not signed by the trusted CA.
[BLOCKED] The leader refuses to establish a session. Attack failed.

○ (env) PS D:\My Projects\Capstone Project> █
```



Results

• (env) PS D:\My Projects\Capstone Project> python demo.py mitm off

```
=====
SCENARIO 2: Man-in-the-middle attack (identity checks OFF)
=====
```

Leader tries to reach follower-1 -- but the address is the attacker's:

```
- [leader->follower-1] sent HELLO + nonce (05c5dadf...)
- [leader->follower-1] got follower cert 'mitm' + Kyber public key (1184 B)
- [leader->follower-1] INSECURE mode: skipping follower identity checks
- [leader->follower-1] encapsulated shared secret; derived session key via HKDF
- [leader->follower-1] sent leader cert + ciphertext + signature (mutual authentication)
- [leader->follower-1] secure session established
[!]      attacker established BOTH sessions -- it is now in the middle
[!]      handshake completed THROUGH the attacker (no identity checks)
```

Leader sends commands; watch the attacker read and change them:

```
[!]      STOLEN plaintext from leader: "SET GAP 20m"
[!]      MODIFIED command before forwarding -> "BRAKE (injected by attacker)"
- [follower-1] ACK 'BRAKE (injected by attacker)' by follower-1 (RTT 404.57 ms)
[!]      STOLEN plaintext from leader: "SET GAP 25m"
- [follower-1] ACK 'SET GAP 25m' by follower-1 (RTT 0.89 ms)
[!]      Without authentication, the attacker read every command and injected a fake BRAKE.
```

• (env) PS D:\My Projects\Capstone Project> █



Expected Outcome & Resources

Final Deliverables:

Winning a Hackathon:

- Develop an innovative solution under time constraints, demonstrating creativity and technical skill. Recognition validates the idea's real-world impact.

Publishing a Paper:

- Share research and findings in an academic format to contribute to the field. Enhances credibility and supports future advancements.

Patenting the Idea:

- Secure intellectual property rights to protect the invention. Enables commercialization and partnership opportunities.



Conclusion

- The proposed authentication framework addresses the need for quantum-resistant vehicle authentication in autonomous platoon networks using CRYSTALS-Dilithium.
- The implementation enables secure vehicle identity verification, message authenticity, and integrity through post-quantum digital signatures.
- Experimental validation demonstrates successful signature generation and verification, effectively preventing message forgery and impersonation attacks.
- This work extends the secure communication framework previously developed and establishes a foundation for trusted, quantum-secure autonomous vehicle communication.



Thank you